Yueqing Xuan

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Education

PhD in Computer Science, RMIT University Aug 2022 – Feb 2026

- **Supervisors**: Prof Jeffrey Chan, Prof Mark Sanderson and Dr Kacper Sokol (USI Switzerland).
- **Scholarship**: Recipient of PhD Scholarship funded by the *Australian Research Council (ARC) Centre of Excellence for Automated Decision-Making and Society (ADM+S)*.
- **Research focus**: Explainable AI, Trustworthy AI, Recommender Systems, Counterfactual Explanations.
- **Thesis**: Accounting for User Diversity in the Pursuit of Fair and Explainable Machine Learning.

Master of Information Technology, University of Melbourne Feb 2020 – Jul 2022

- **Specialisation**: Artificial Intelligence.
- **Supervisors**: Dr Lea Frermann and Dr Adi Tarigoppula.
- **WAM**: 84.50 – H1, First Class Honours.
- **Achievements**: Recognised on the *Dean's Honours List 2021* at the University of Melbourne for top 5% of students. Awarded *Melbourne Graduate Scholarship 2022*.

Bachelor of Commerce, University of New South Wales Jul 2015 – Jul 2018

- **Major**: Accounting and Information Systems.

Publications

Conference and Journal Papers

[1] **Yueqing Xuan**, Kacper Sokol, Danula Hettiachchi. "Explanation Navigator: Rectifying Out-of-Scope Human Interpretations of Leaky AI Explanations through Conversational Guidance". In: arXiv preprint arXiv:2609.22707. 2026.

[2] **Yueqing Xuan**, Kacper Sokol, Mark Sanderson, Jeffrey Chan. "Diversity-Augmented Negative Sampling for Implicit Collaborative Filtering". Proceedings of the ACM Web Conference 2026 (WWW'26), 2026, https://doi.org/10.1145/3774904.3792346.

[3] **Yueqing Xuan**, Kacper Sokol, Mark Sanderson, Jeffrey Chan. "Evaluating and Addressing Fairness Across User Groups in Negative Sampling for Recommender Systems". Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM), 2025, https://doi.org/10.1145/3746252.3761263.

[4] **Yueqing Xuan**, Kacper Sokol, Mark Sanderson, Jeffrey Chan. "Perfect Counterfactuals in Imperfect Worlds: Modelling Noisy Implementation of Actions in Sequential Algorithmic Recourse". Machine Learning, 114, no.8 (2025): 187, https://doi.org/10.1007/s10994-025-06821-1.

[5] Kacper Sokol*, Edward Small*, **Yueqing Xuan***. "Navigating Explanatory Multiverse through Counterfactual Path Geometry". Machine Learning, 114, no.6 (2025): 1-33, https://doi.org/10.1007/s10994-025-06769-2. *Equal contribution.

[6] **Yueqing Xuan**, Kacper Sokol, Mark Sanderson, Jeffrey Chan. "Leveraging Complementary AI Explanations to Mitigate Misunderstanding in XAI". 2025 IEEE Swiss Conference on Data Science (SDS), 2025, https://doi.org/10.1109/SDS66131.2025.00034.

[7] **Yueqing Xuan**, Edward Small, Kacper Sokol, Danula Hettiachchi, Mark Sanderson. "Comprehension

Is a Double-Edged Sword: Over-Interpreting Unspecified Information in Intelligible Machine Learning Explanations". International Journal of Human–Computer Studies (IJHCS), 2024, 103376, ISSN 1071-5819, https://doi.org/10.1016/j.ijhcs.2024.103376.

Workshop Papers

[1] Edward Small*, **Yueqing Xuan***, Danula Hettiachchi*, Kacper Sokol*. "Helpful, Misleading or Confusing: How Humans Perceive Fundamental Building Blocks of Artificial Intelligence Explanations". ACM CHI Workshop on Human-Centered Explainable AI (HCXAI). In: arXiv preprint arXiv:2303.00934. 2023.

[2] Kacper Sokol*, Edward Small*, **Yueqing Xuan***. "Navigating Explanatory Multiverse through Counterfactual Path Geometry". ICML (International Conference on Machine Learning) Workshop on Counterfactuals in Minds and Machines. In: arXiv preprint arXiv:2306.02786. 2023.

Theses

[1] **Yueqing Xuan**. "Accounting for User Diversity in the Pursuit of Fair and Explainable Machine Learning". PhD thesis. School of Computing Technologies, RMIT University, 2026.

[2] **Yueqing Xuan**. "Multi-modal Entity Linker". Master's thesis. School of Engineering and Information Technology, University of Melbourne, 2021.

Academic Experience

Research Fellow, RMIT University, Melbourne Mar 2026 – Present

- Conduct research on misunderstanding of AI explanations, focusing on follow-up interactions, conversational explanation systems, and multimodal explainers.

Tutor, RMIT University, Melbourne Feb 2024 – Nov 2025

- Tutored *COSC1285 Algorithms and Analysis* (1st-year M.Sc., Semesters 1–2, 2025) and *COSC2123 Algorithms and Analysis* (2nd-year B.Sc., Semesters 1–2, 2024).
- Led weekly workshops and tutorials for classes of approximately 40 students, explaining algorithmic concepts, demonstrating worked examples, and guiding problem-solving activities using Python.
- Marked assignments and examinations and provided constructive feedback on algorithmic reasoning, implementation, and problem-solving approaches.
- Conducted student consultations and provided individual support on course concepts, programming tasks, and assessments.

Research Assistant, University of Melbourne, Melbourne Nov 2021

- Research project: Humanoid robotic arm control through verbal commands (GitHub Link).
- Integrated the Porcupine wake-word engine and IBM Watson speech recognition system into a robotic arm control framework, enabling real-time interaction with an anthropomorphic robotic limb through natural-language commands.

Professional Activities

Academic Service

- Paper reviewer for DMKD (2026), WSDM (2026), SIGIR (2025), and SIGIR (2023).
- Organiser, RMIT Recommender and Responsible AI (R2AI) Reading Group.